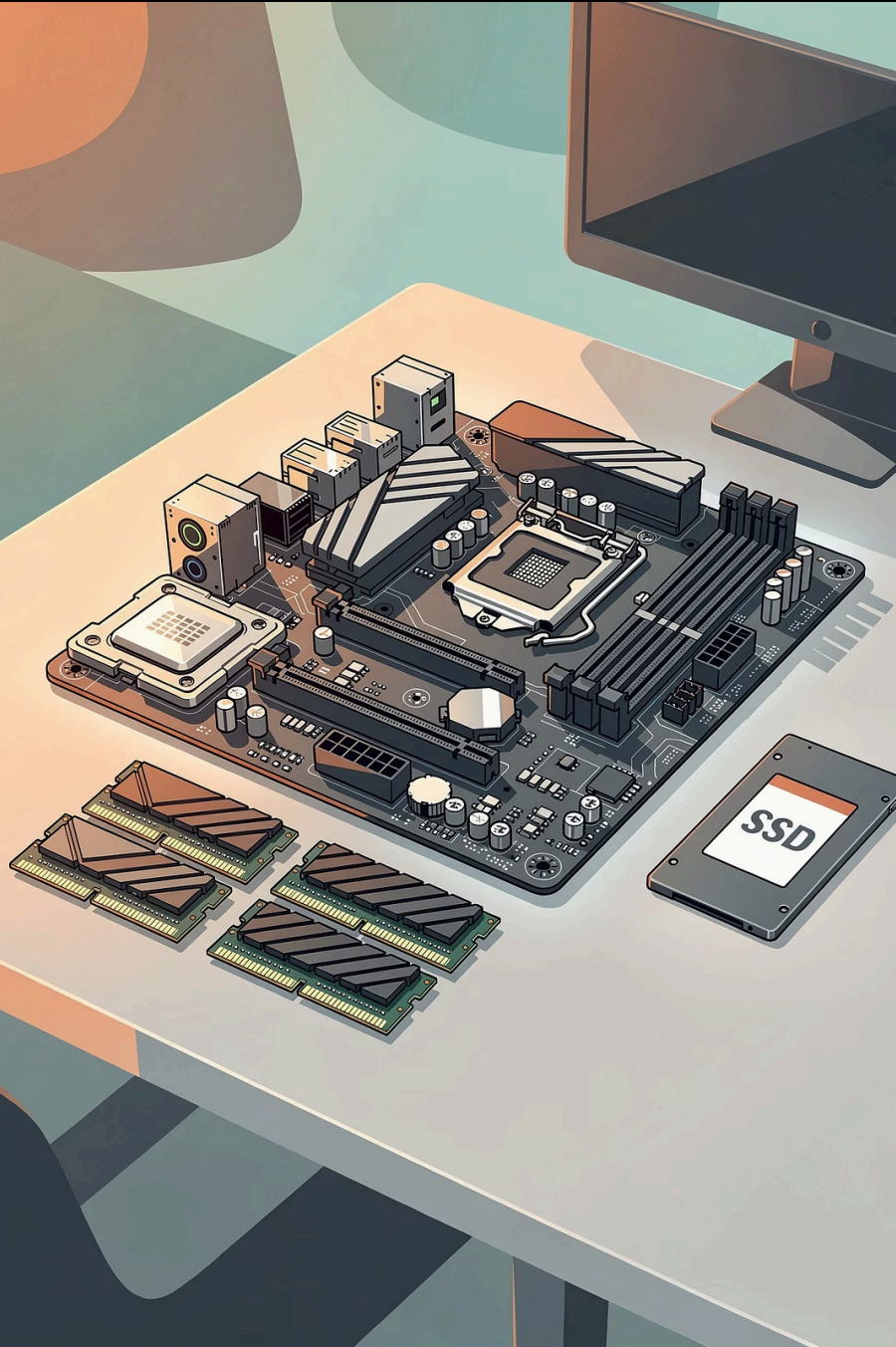




Computer Hardware & Operating Systems

A structured overview of the most important hardware components, how a computer works, and the basics of operating Windows.

FUNDAMENTALS

IT SKILLS

A computer works like a team

Each component has a specific task – only together do they become a usable system. The four basic areas are particularly important:



Processor

Computing and controlling



RAM

Keeping active data ready



Drives

Permanent storage



Peripherals

Input and output

The Processor (CPU) – The Computer's Brain

The processor processes instructions, executes calculations, and controls many operations. It doesn't "decide for itself" what is sensible, but it **executes commands extremely quickly and in the correct order**.

Cores

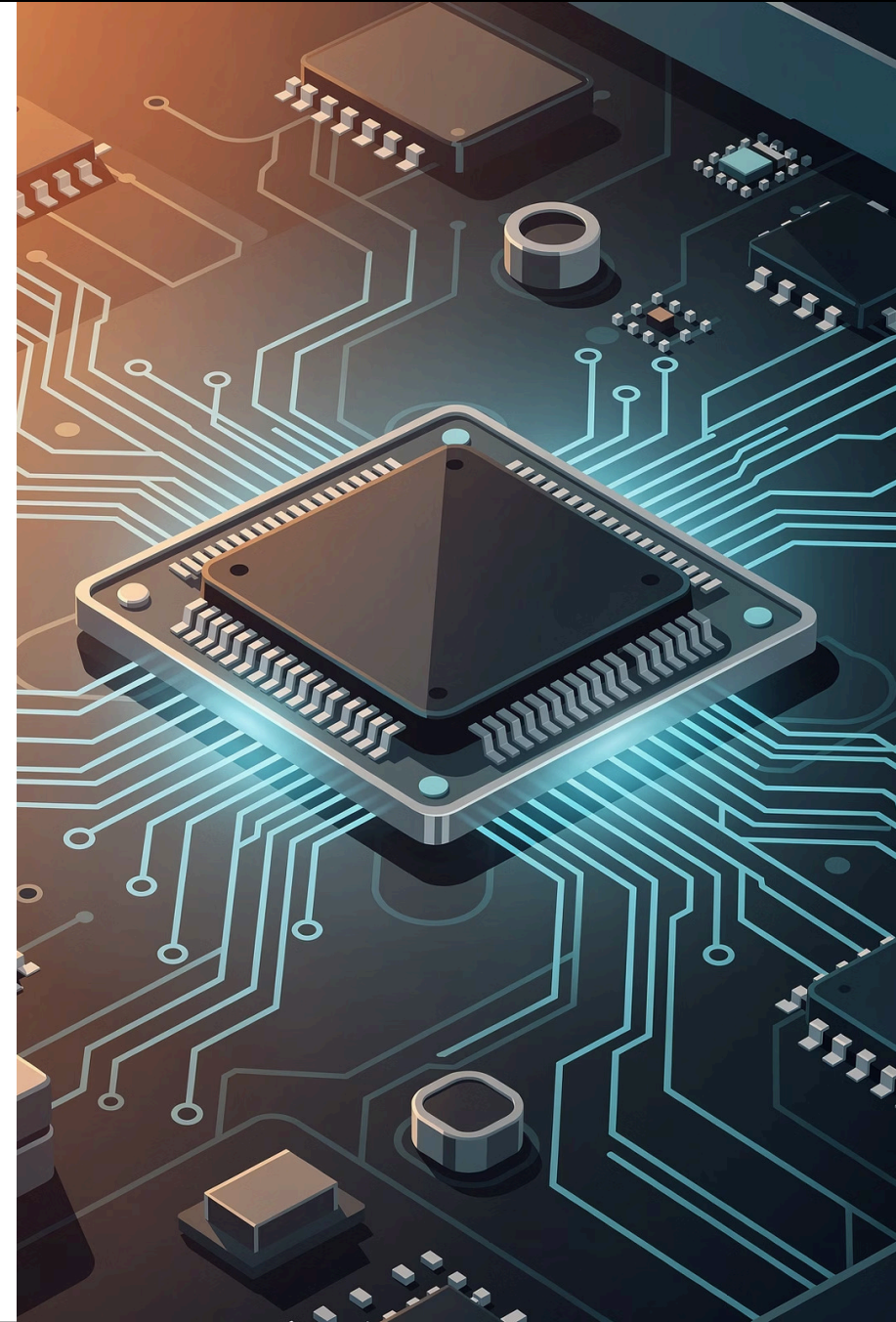
More cores allow for better parallel processing of multiple tasks.

GHz

Clock speed is only one part of performance – architecture and cache also matter.

Comparison

The processor is like a **chef in the kitchen** – it needs ingredients close at hand (RAM).



The Memory (RAM) – the workspace

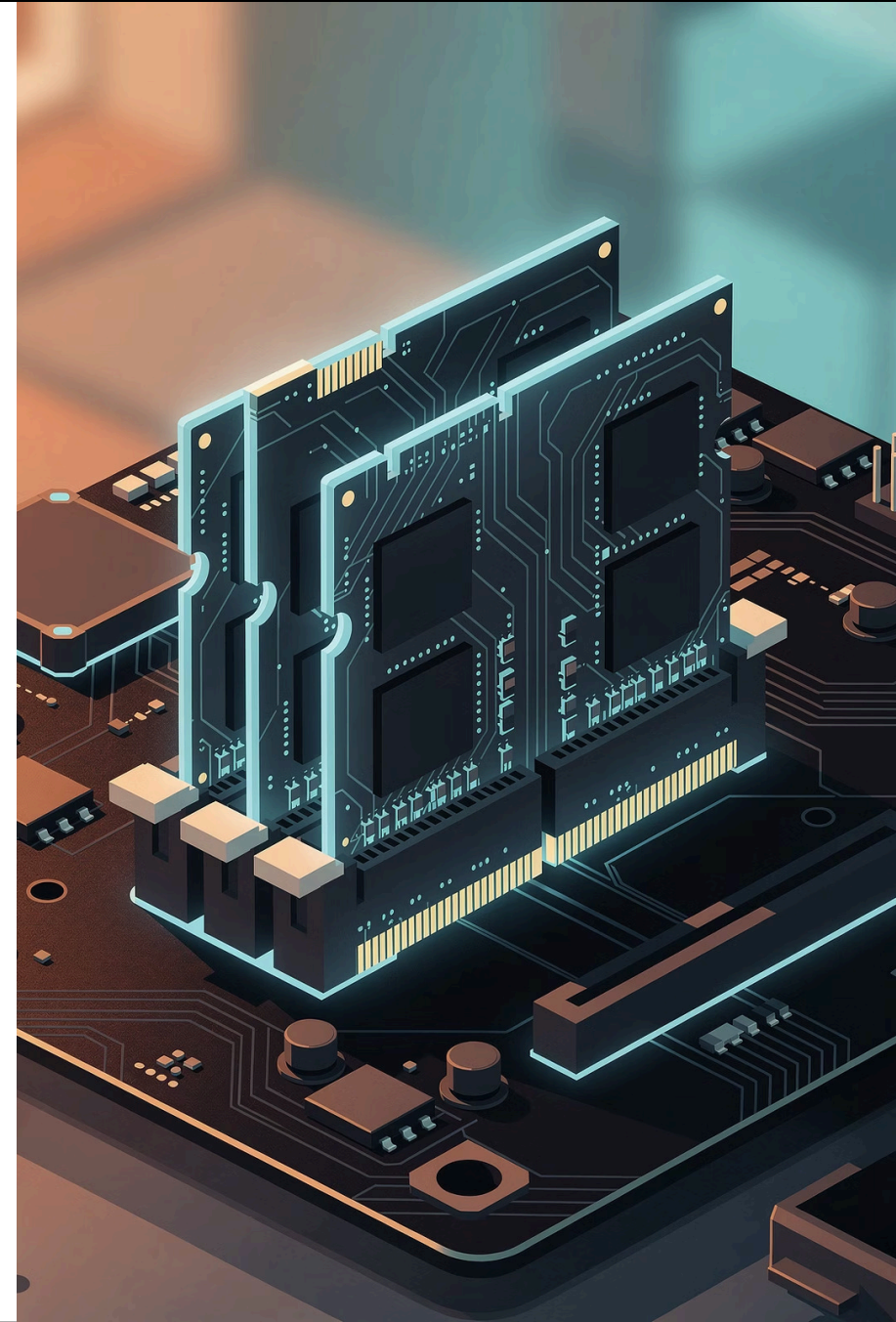
RAM is the **short-term memory** of the computer. It holds data and programs that are currently in use so that the processor can access them quickly. **RAM is fast, but volatile** – its contents are lost when the computer is turned off.

Why RAM is important

When you have a browser, word processor, music, and video call open at the same time, all that active data needs to be readily available.

The desk analogy

RAM is like the **workspace on your desk**. If it's too small, you constantly have to clear things away and retrieve them again – everything slows down.



Drives – permanent storage

Drives store data **permanently**: operating systems, programs, documents, images, videos. Today there are two important types:

SSD – Solid-State-Drive

Stores data electronically, no moving parts. **Significantly faster**, quieter, and more robust. Windows starts faster, programs open more smoothly.

HDD – Hard Disk Drive

Stores on rotating magnetic disks. **Cheaper per gigabyte**, but slower and mechanically more sensitive.

- Mnemonic: The drive is like a **filing cabinet** – everything remains preserved, even when you finish your work. The RAM is just the current workspace.



Peripherals – The Connection to the Outside World

Peripheral devices enable the input and output of data. Without them, the computer could perform calculations internally, but you would hardly be able to work with it in a meaningful way.



Input Devices

Keyboard, mouse, microphone, webcam – you enter data and commands into the computer.



Output Devices

Monitor, printer, speakers, headphones – the computer returns results to you.



Storage Devices

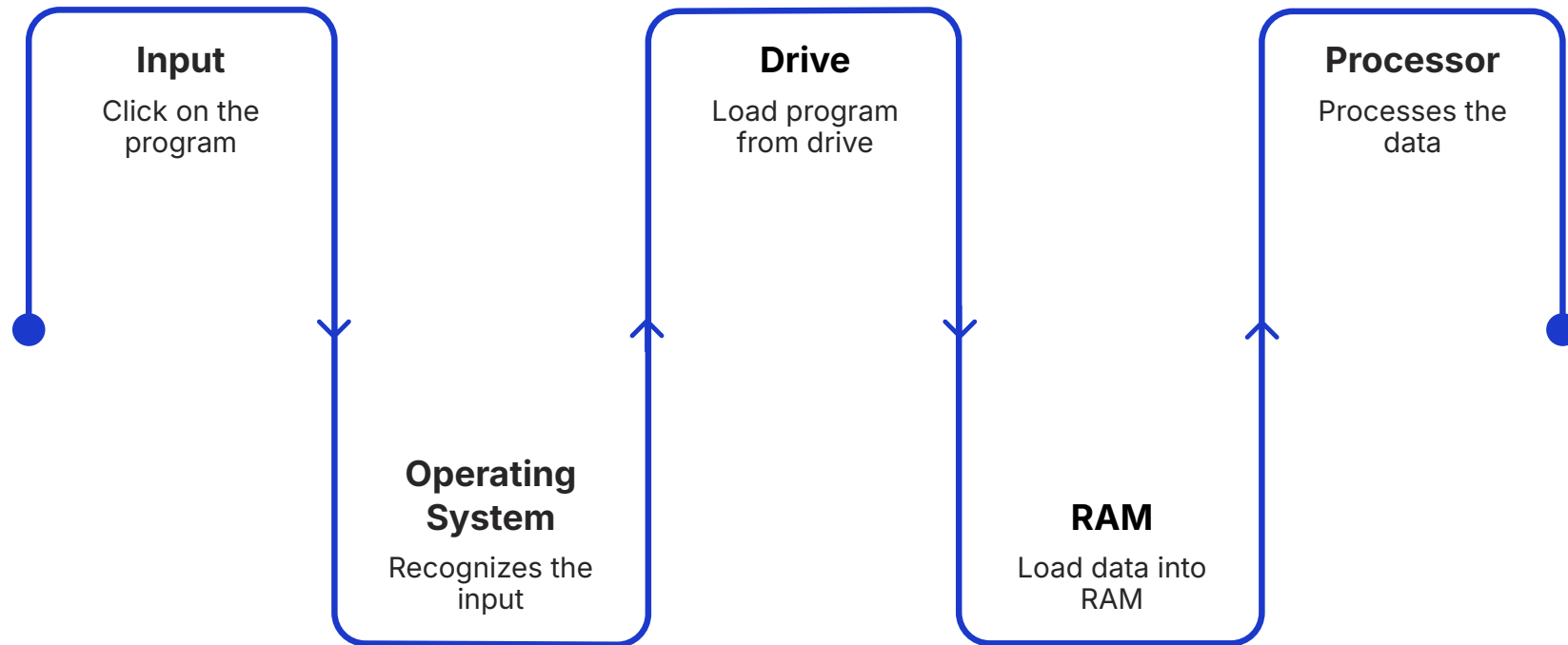
USB drive or external hard drive – additional storage, also permanent depending on the device.

Components Comparison

Component	Main Task	Storage Duration	Simple Analogy
Processor (CPU)	Calculate and Control	No storage	Brain / Chef
Memory (RAM)	Keep active data ready	Only with power	Desk
Drive (SSD/HDD)	Store data permanently	Even without power	Filing cabinet
Peripherals	Input and Output	Varies	Connection to the outside world

Basic Functionality – Step by Step

A computer does not work through a single component, but through **interaction in a chain**. Here is the simplified process using the example "opening a program":



This is the basic idea behind almost everything that happens on a computer – from input to output, every action goes through this chain.

What happens when you turn it on

01

Power supply

The hardware is powered up.

02

BIOS / UEFI

Basic firmware starts, checks existing devices, and initiates the boot process.

03

Loading the operating system

The operating system is read from the drive and loaded into the main memory.

04

Desktop appears

Only then does the login screen or desktop appear.





CHAPTER 2

The Operating System

Hardware alone would be difficult for most people to use. The **operating system** is the fundamental software that manages hardware, launches programs, and provides a user interface – using **Windows** as an example.

What an Operating System does

The operating system is the **organizer and mediator** of the computer. It turns many technical individual parts into a usable work environment.

→ **Manage hardware**

It detects and controls all connected devices.

→ **Launch programs**

It loads applications and ensures that several can run at the same time.

→ **Manage files**

It organizes files and folders on the drives.

→ **Provide interface**

It displays an interface you can work with – desktop, taskbar, start menu.

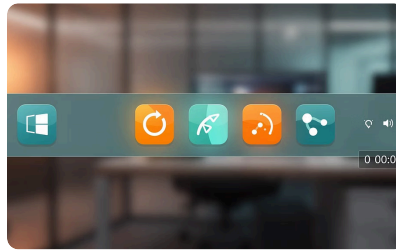
Desktop, Taskbar & Start Menu

These three elements are the **center of operation in Windows**. Anyone who understands them will find their way around the system very easily.



Desktop

The large area after logging in. Contains the background, shortcuts, folders, and the Recycle Bin. Icons are often just **shortcuts**, not the actual program.



Taskbar

Bar at the bottom of the screen. Contains the Start button, open programs, network, volume, and time. Allows for quick switching between windows.



Start Menu

The **central gateway** to Windows. Opens via the Start button or the Windows key. Contains programs, search, settings, and power options.

Window Management – Basics

As soon as you open programs, you work with **windows**. Three buttons in the top right corner control every window:

Minimize

The window disappears from view, but remains open and can be accessed in the taskbar.

Maximize

The window fills almost the entire screen.

Close

The window or program is terminated.

Additionally, you can **move** windows (drag the title bar) and **resize** them (drag the edge or corner). The active window is always in the foreground.

Switching Between Windows & Navigation

Switching Between Windows

- Click on the program icon in the taskbar
- **Alt + Tab** for quick switching
- Directly clicking on a visible window

Basic Mouse Actions

- **Single-click** – select
- **Double-click** – open
- **Right-click** – context menu
- **Scrolling** – through lists and pages
- **Drag & Drop** – move

- ❏ The **File Explorer** is the most important tool for navigating through folders, drives, and files. Anyone "searching for something on the computer" will very often end up here.

Why window management is so important

Many beginners underestimate this area – but it is the foundation for much of the confidence you feel when using computers. Understanding what is open, what is merely minimized, what is active, and how to close programs purposefully means you will no longer feel "lost" in the system.

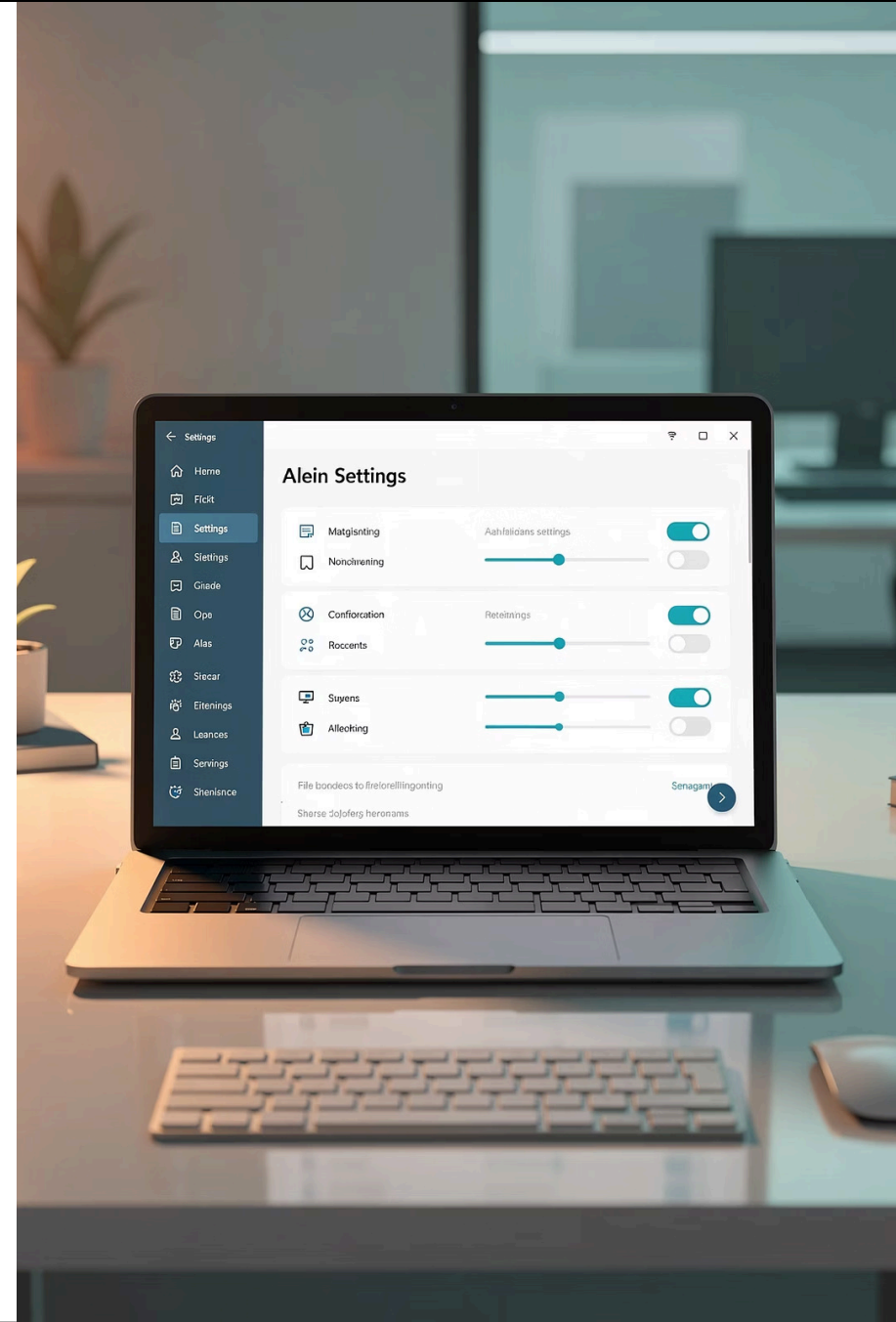
- ❑ Window management is one of the essential **basic skills** for using a computer.



CHAPTER 3

Elementary System Settings

Beyond daily operation, you should be able to adjust basic settings. These adapt the computer to **your language, your time zone, and your display**. In Windows, you can find most of these in the **Settings** app.



Time, Language & Display Options

Time & Date

The correct time affects file timestamps, calendars, emails, and security functions. Adjustable: automatic synchronization, time zone, time format.

Language & Keyboard

Display language (menus and system text) and **keyboard layout** (which key produces which character) are separate settings – especially important on shared devices.

Display Options

Resolution determines screen sharpness. **Scaling** enlarges text and icons. If everything seems too small, scaling is often the main issue – not the resolution.

Sign In, Sign Out, Shut Down, Restart

These functions are often confused – it is worth distinguishing them clearly:

Action	What happens?	Computer on?	Typical purpose
Sign In	User session & profile are loaded	Yes	Work with your own account
Sign Out	Session ends, computer stays on	Yes	Switch users, clean exit
Shut Down	System closes, device powers off	No	Finish work, save energy
Restart	System shuts down and starts up again	Briefly no	Updates, troubleshooting

Where to find these functions

Start Menu

- **User icon** → Sign in or sign out
- **Power icon** → Shut down or restart

Other ways

- Sign-in screen
- Right-click on the Start button
- Keyboard shortcuts

Why this matters

Confusing "Sign out" with "Close window" risks data loss. Not understanding the difference between "Shut down" and "Restart" leads to confusion about why updates only take effect after a restart.

These seemingly simple functions are truly the **foundations of secure computer usage**.